# ABANYERONDO: Community Security Availability and Emergency Assistance System

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## Problem Statement

Community patrols (Irondo) support local security in Rwanda, but a citizen may not know which official duty contact serves a location or is actually available. This is especially challenging when a citizen travels beyond a saved home area. Research on locating ambulance emergencies in Rwanda illustrates the importance of clear location information, although it does not directly study Irondo (Hunt et al., 2024).

## Proposed Solution

Develop an Oracle Database/PL/SQL-based availability platform that receives authenticated hiring, transfer, dismissal and suspension status events from a separate simulated personnel authority; it never performs those employment decisions or payroll. Events update assignment and eligibility history. Authorized duty check-ins confirm actual availability. An Oracle APEX public interface searches by saved home location, alternative selected location or permitted current location, returning only approved duty contacts and displaying a freshness indicator and official emergency fallback.

## Objectives

Main objective: improve citizens' access to verified local security duty contacts. Specific objectives: (1) maintain normalized, sourced Rwanda location records; (2) validate, synchronize and audit external status events; (3) enforce confirmed duty-availability and public-contact access rules; (4) provide a live Oracle-connected citizen search interface.

## Data Source

The imported real-world dataset is the openly licensed Open Admin Data Rwanda administrative divisions repository (CC BY 4.0). We extracted and validated 451 non-personal records: 5 province-level units, 30 districts and 416 sectors, with zero duplicate IDs and zero broken parent references. The World Bank 2012 boundaries and NISR 2022 services provide research context, but were not imported into this package. This phase has no cell/village data; all personnel, contacts, duty events and incidents for demonstration will be synthetic and labeled as such.

## Planned Tables

SOURCE_IMPORT, PROVINCE, DISTRICT, SECTOR, OFFICER_REFERENCE, PERSONNEL_EVENT, OFFICER_ASSIGNMENT, DUTY_SHIFT, DUTY_CHECKIN, DUTY_CONTACT, CITIZEN_LOCATION_PREF, CONTACT_LOOKUP_AUDIT, SYNC_AUDIT. Primary/foreign keys, historical effective dates, normalization to 3NF and ERD will be refined in Phase II.

## Planned PL/SQL Features

Packages for idempotent event ingestion and availability lookup; procedures to validate external updates; functions to determine eligibility and freshness; cursors to return eligible contacts; triggers, exceptions, constraints and audit logs to reject contradictory or duplicate events and protect contact visibility. A scheduled monitor can flag stale upstream status.

## Innovation

A live Oracle APEX application linked to Oracle PL/SQL provides saved-location and alternative-area lookup, duty-contact visibility and a supervisor event-activity screen. A simulated upstream personnel event (e.g., suspension) must change live public results immediately without editing HR records in the app.

## References (APA)

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